

WEIPA, Australia

WMO: 941700

Lat: 12.6778S

Lon: 141.9208E

Elev: 62

StdP: 14.66

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	62.2	64.0	49.7	53.0	86.4	52.8	59.6	84.6	21.9	84.4	20.2	84.0	2.8	130	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	19.7	96.4	72.7	94.9	72.8	93.3	73.2	80.7	86.9	80.0	86.2	79.6	85.6	11.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	151.3	83.1	78.4	148.0	82.5	78.0	145.9	82.1	44.4	87.1	43.8	86.7	43.2	85.9	97.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.8	17.0	15.6	DB	57.7	99.5	3.0	1.2	55.5	100.3	53.8	101.1	52.1	101.7	49.9	102.6
			WB	55.1	84.2	2.8	4.7	53.0	87.6	51.4	90.4	49.8	93.0	47.8	96.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	82.0	81.6	81.6	81.1	79.4	77.3	76.2	77.1	80.0	82.9	84.3	83.9	(6)
(7)		DBStd	3.33	2.11	2.02	1.74	1.67	2.04	2.41	2.49	2.36	2.29	2.21	1.81	2.05	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11172	992	886	979	933	911	818	814	839	900	1020	1030	1050	(10)
(11)		CDD65	5697	527	466	514	483	446	368	349	374	450	555	580	585	(11)
(12)		CDH74	55347	5155	4388	4739	4591	3963	2895	2785	3317	4506	6054	6580	6374	(12)
(13)		CDH80	21792	1755	1387	1546	1685	1398	909	869	1257	2084	2956	3173	2773	(13)
(14)	Wind	WSAvg	7.0	5.9	5.5	5.3	6.9	7.7	8.0	8.3	7.7	7.8	7.8	6.7	5.8	(14)
(15)	Precipitation	PrecAvg	74.1	18.5	19.4	15.7	4.1	0.6	0.1	0.0	0.0	0.2	0.9	4.3	10.2	(15)
(16)		PrecMax	105.9	43.8	36.2	41.9	13.9	5.1	1.3	0.4	0.4	1.3	4.6	17.8	20.9	(16)
(17)		PrecMin	47.6	7.2	6.3	2.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.6	(17)
(18)		PrecStd	15.4	7.8	7.8	7.8	3.7	0.9	0.2	0.1	0.1	0.2	1.1	3.5	4.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.8	92.2	91.7	91.7	90.9	89.8	89.5	91.8	95.8	98.0	98.3	97.4	(19)
(20)			MCWB	76.6	77.8	77.4	76.5	75.2	72.3	70.4	69.3	69.7	72.0	72.3	74.8	(20)
(21)		2%	DB	91.2	90.3	90.4	90.4	89.3	88.1	87.8	89.9	93.7	96.0	96.5	94.9	(21)
(22)			MCWB	77.8	78.3	77.3	75.9	74.1	72.0	70.2	69.3	70.1	72.0	72.7	75.3	(22)
(23)		5%	DB	89.5	88.8	89.0	89.2	88.1	86.7	86.4	88.3	92.0	94.3	94.9	92.9	(23)
(24)			MCWB	78.2	78.6	77.5	75.4	73.6	71.7	69.6	69.1	70.1	71.7	73.0	76.2	(24)
(25)		10%	DB	87.9	87.3	87.5	88.0	86.8	85.3	84.9	86.7	90.2	92.4	92.8	91.0	(25)
(26)			MCWB	78.6	78.7	77.8	75.4	73.3	71.3	69.3	69.0	70.1	71.8	74.0	77.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	81.0	81.0	80.0	78.4	76.8	75.2	75.4	76.8	79.2	80.4	81.0	(27)
(28)			MCDWB	87.2	87.1	86.8	85.8	85.5	83.4	83.0	83.6	84.8	86.3	87.8	87.6	(28)
(29)		2%	WB	80.5	80.1	80.0	79.0	77.2	75.0	73.7	73.8	75.5	77.5	79.2	80.1	(29)
(30)			MCDWB	86.4	86.0	85.9	85.2	84.3	82.9	81.9	82.7	84.0	85.1	86.4	87.0	(30)
(31)		5%	WB	79.9	79.7	79.3	78.2	76.2	73.8	72.5	72.7	74.4	76.5	78.4	79.5	(31)
(32)			MCDWB	85.6	85.3	85.0	84.3	83.5	82.0	81.3	82.0	83.5	84.7	85.6	86.3	(32)
(33)		10%	WB	79.2	79.1	78.8	77.5	75.3	72.9	71.4	71.5	73.4	75.6	77.7	78.9	(33)
(34)			MCDWB	84.8	84.4	84.3	83.4	82.7	81.3	80.7	81.3	83.0	84.5	85.0	85.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.0	11.3	12.5	15.0	16.6	17.6	19.4	21.8	23.3	22.3	19.7	15.8	(35)
(36)			MCDBR	15.0	13.9	14.9	16.7	17.9	19.0	20.8	23.2	25.0	24.0	22.5	19.0	(36)
(37)			MCWBR	4.6	4.5	4.7	5.0	5.6	5.8	6.3	7.0	7.1	6.4	6.1	5.4	(37)
(38)		5% WB	MCDWB	12.3	11.8	12.4	13.5	14.7	16.3	17.4	20.1	22.3	20.9	17.8	15.2	(38)
(39)	MCWBR		4.8	4.6	4.8	4.9	5.4	6.1	6.2	7.2	7.2	6.4	5.6	5.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.409	0.406	0.392	0.389	0.376	0.369	0.360	0.374	0.395	0.407	0.418	0.409	(40)	
(41)		taud	2.519	2.528	2.570	2.555	2.567	2.574	2.588	2.528	2.452	2.426	2.424	2.489	(41)	
(42)		Ebn at Noon	297	295	294	284	278	275	281	285	288	291	292	296	(42)	
(43)		Edh at Noon	36	35	33	32	30	29	29	32	36	39	39	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1619	1636	1644	1703	1565	1434	1514	1765	2032	2193	2170	1932	(44)	
(45)		RadStd	195	262	129	143	92	98	82	103	85	107	126	210	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	+202	+196
(47) Regional (3 neighbors)		N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	+202	+196

Nomenclature: See separate page